Data Privacy: Homeworks #4 Solutions

1. **Neyman–Pearson Lemma (Detailed Proof).** Let T be a random variable taking values in a measurable space $(\mathcal{T}, \mathcal{B})$. Consider two simple hypotheses

$$H_0 : T \sim p_0, \qquad H_1 : T \sim p_1,$$

where p_0, p_1 are densities with respect to a common dominating measure μ. A (randomized) test is a measurable function $\phi : \mathcal{T} \to [0, 1]$, where $\phi(t)$ is the probability of rejecting H_0 after observing $T = t$. Define

$$\alpha(\phi) = \mathbb{E}_{H_0}[\phi(T)] \quad \text{(size, type-I error)}, \qquad \beta(\phi) = \mathbb{E}_{H_1}[\phi(T)] \quad \text{(power)}.$$

Fix a level $\alpha_0 \in (0, 1)$. Prove the following three claims.

(a) **Existence.** Show that there exist a threshold $\tau \geq 0$ and a randomization $\gamma \in [0, 1]$ such that the test

$$\phi^*(t) = \begin{cases} 1 & \text{if } p_1(t) > \tau\, p_0(t), \\ \gamma & \text{if } p_1(t) = \tau\, p_0(t), \\ 0 & \text{if } p_1(t) < \tau\, p_0(t), \end{cases}$$

satisfies $\alpha(\phi^*) = \alpha_0$ exactly.
Hint: Consider the function $G(\tau) = \Pr_{H_0}[\Lambda(T) > \tau]$ where $\Lambda(t) = p_1(t)/p_0(t)$. Show G is right-continuous and non-increasing, then pick τ so that $G(\tau) \leq \alpha_0 \leq G(\tau-)$, and choose γ to hit equality.

(b) **Optimality.** Show that for every test ϕ with $\alpha(\phi) \leq \alpha_0$, we have $\beta(\phi) \leq \beta(\phi^*)$.
Hint: Verify that for all t, $(\phi^*(t) - \phi(t))\,(p_1(t) - \tau\, p_0(t)) \geq 0$, then integrate.

(c) **Uniqueness.** Suppose ϕ is any most-powerful level-α_0 test. Show that on the set $\{t : p_1(t) \neq \tau\, p_0(t)\}$ we have $\phi(t) = \phi^*(t)$ for μ-almost every t.
Hint: The integrand in part (b) is non-negative and integrates to 0.

Solution:

(a) **Existence of (τ, γ).** Define the likelihood ratio $\Lambda(t) = p_1(t)/p_0(t)$, with the convention $\Lambda = +\infty$ on $\{p_0 = 0, p_1 > 0\}$. Consider the survival function

$$G(\tau) = \Pr_{H_0}[\Lambda(T) > \tau], \quad \tau \in [0, \infty).$$

The sets $\{\Lambda > \tau\}$ are nested-decreasing in τ, so G is non-increasing. As $\tau_n \downarrow \tau$, $\{\Lambda > \tau_n\} \uparrow \{\Lambda > \tau\}$, so by monotone convergence $G(\tau_n) \to G(\tau)$, i.e., G is right-continuous. The left limit is $G(\tau-) = \Pr_{H_0}[\Lambda \geq \tau]$, and $G(\tau-) - G(\tau) = \Pr_{H_0}[\Lambda = \tau]$.
Set $\tau^* = \inf\{s \geq 0 : G(s) \leq \alpha_0\}$. By right-continuity and monotonicity, $G(\tau^*) \leq \alpha_0 \leq G(\tau^*-)$. If $G(\tau^*) = G(\tau^*-)$ then $\alpha_0 = G(\tau^*)$ and any γ works (take $\gamma = 0$). Otherwise set

$$\gamma = \frac{\alpha_0 - G(\tau^*)}{G(\tau^*-) - G(\tau^*)} \in [0, 1].$$

Then

$$\alpha(\phi^*) = \Pr_{H_0}[\Lambda > \tau^*] + \gamma \Pr_{H_0}[\Lambda = \tau^*] = G(\tau^*) + \gamma(G(\tau^*-) - G(\tau^*)) = \alpha_0.$$

(b) **Optimality.** We claim the pointwise inequality $(\phi^*(t)-\phi(t))(p_1(t)-\tau^* p_0(t)) \geq 0$:

- If $p_1(t) > \tau^* p_0(t)$: $\phi^*(t) = 1 \geq \phi(t)$, second factor positive — product ≥ 0.
- If $p_1(t) < \tau^* p_0(t)$: $\phi^*(t) = 0 \leq \phi(t)$, second factor negative — product ≥ 0.
- If $p_1(t) = \tau^* p_0(t)$: second factor is 0.

Integrate against μ:

$$0 \leq \int (\phi^* - \phi)(p_1 - \tau^* p_0)\, d\mu = (\beta(\phi^*) - \beta(\phi)) - \tau^*(\alpha(\phi^*) - \alpha(\phi)).$$

Since $\alpha(\phi^*) = \alpha_0 \geq \alpha(\phi)$ and $\tau^* \geq 0$, the second term is non-negative, so $\beta(\phi^*) \geq \beta(\phi)$.

(c) **Uniqueness.** If ϕ is also most powerful, $\beta(\phi) = \beta(\phi^*)$ and $\alpha(\phi) \leq \alpha_0$, so

$$0 \leq \int (\phi^* - \phi)(p_1 - \tau^* p_0)\, d\mu = -\tau^*(\alpha_0 - \alpha(\phi)) \leq 0.$$

The integral equals 0. The integrand is ≥ 0 pointwise, so it vanishes μ-a.e. On $\{p_1 > \tau^* p_0\}$, $p_1 - \tau^* p_0 > 0$ forces $\phi = \phi^* = 1$ a.e.; on $\{p_1 < \tau^* p_0\}$, $\phi = \phi^* = 0$ a.e. Hence $\phi = \phi^*$ μ-a.e. on $\{p_1 \neq \tau^* p_0\}$.

2. **DP $\Rightarrow$ MIA Bounds.** Let $M : \mathcal{X}^n \to \mathcal{Y}$ be (ε, δ)-differentially private, and let D_0, D_1 be neighboring datasets. Consider any (possibly randomized) binary test $\mathcal{A} : \mathcal{Y} \to \{0, 1\}$ that tries to distinguish $M(D_0)$ from $M(D_1)$. Write

$$\mathrm{FPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_0)) = 1], \qquad \mathrm{TPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_1)) = 1].$$

(a) Show that

$$\mathrm{TPR}(\mathcal{A}) \leq e^{\varepsilon}\, \mathrm{FPR}(\mathcal{A}) + \delta, \qquad 1 - \mathrm{FPR}(\mathcal{A}) \leq e^{\varepsilon}\,(1 - \mathrm{TPR}(\mathcal{A})) + \delta.$$

(b) Define the MI advantage $\mathrm{Adv}(\mathcal{A}) = \mathrm{TPR}(\mathcal{A}) - \mathrm{FPR}(\mathcal{A})$. Show the loose bound

$$\mathrm{Adv}(\mathcal{A}) \leq e^{\varepsilon} - 1 + \delta.$$

Then, by combining both inequalities of part (a), derive the tighter Kairouz–Oh–Viswanath bound

$$\mathrm{Adv}(\mathcal{A}) \leq \frac{e^{\varepsilon} - 1 + 2\delta}{e^{\varepsilon} + 1}.$$

(c) Conclude that for $\varepsilon = 0.1$ and $\delta = 10^{-5}$, no adversary can exceed advantage ≈ 0.105, but for $\varepsilon = 8$ the bound is vacuous.

Solution:

(a) Let $S = \{y : \mathcal{A}(y) = 1\}$ (averaged over $\mathcal{A}$'s internal randomness if randomized). Apply (ε, δ)-DP to S with the ordered pair (D_1, D_0):

$$\Pr[M(D_1) \in S] \le e^{\varepsilon} \Pr[M(D_0) \in S] + \delta \iff \mathrm{TPR} \le e^{\varepsilon}\mathrm{FPR} + \delta.$$

Apply it to S^c with (D_0, D_1):

$$\Pr[M(D_0) \in S^c] \le e^{\varepsilon} \Pr[M(D_1) \in S^c] + \delta \iff 1 - \mathrm{FPR} \le e^{\varepsilon}(1 - \mathrm{TPR}) + \delta.$$

(b) **Loose bound.** From the first inequality and $\mathrm{FPR} \le 1$,

$$\mathrm{Adv} = \mathrm{TPR} - \mathrm{FPR} \le (e^{\varepsilon} - 1)\mathrm{FPR} + \delta \le e^{\varepsilon} - 1 + \delta.$$

Tighter (Kairouz–Oh–Viswanath). The two inequalities define a polygonal feasible region; the maximum of $\mathrm{TPR} - \mathrm{FPR}$ occurs at the vertex where both are tight. Rewriting the second as

$$\mathrm{TPR} \ge 1 - e^{-\varepsilon}(1 - \mathrm{FPR}) - \delta e^{-\varepsilon},$$

and setting equal to $e^{\varepsilon}\mathrm{FPR} + \delta$, we solve

$$(e^{\varepsilon} - e^{-\varepsilon})\mathrm{FPR} = (1 - e^{-\varepsilon})(1 - \delta) \implies \mathrm{FPR}^{\star} = \frac{1 - \delta}{e^{\varepsilon} + 1}.$$

Then $\mathrm{TPR}^{\star} = e^{\varepsilon}\mathrm{FPR}^{\star} + \delta = \frac{e^{\varepsilon}+\delta}{e^{\varepsilon}+1}$, so

$$\mathrm{Adv}^{\star} = \mathrm{TPR}^{\star} - \mathrm{FPR}^{\star} = \frac{e^{\varepsilon} + \delta - (1 - \delta)}{e^{\varepsilon} + 1} = \frac{e^{\varepsilon} - 1 + 2\delta}{e^{\varepsilon} + 1}.$$

(c) For $\varepsilon = 0.1, \delta = 10^{-5}$: loose bound $= e^{0.1} - 1 + 10^{-5} \approx 0.1052$; tighter bound ≈ 0.0500. For $\varepsilon = 8$: $e^{8} \approx 2981$, so loose bound is ≈ 2980 and tighter bound is ≈ 1. Both useless — advantage is automatically ≤ 1.

3. **Bayes-Optimal Membership Test.** Let $B \in \{0, 1\}$ be a Bernoulli(π) membership indicator and let T be an observed statistic with conditional density $p_b(t)$ given $B = b$. Define the loss of any decision rule $\hat{B} : \mathcal{T} \to \{0, 1\}$ by the 0-1 error $\Pr[\hat{B}(T) \ne B]$.

   (a) Compute the posterior $\Pr[B = 1 \mid T = t]$ in terms of p_0, p_1, π.

   (b) Show that the Bayes-optimal rule (the MAP rule) is

   $$\hat{B}_{\mathrm{MAP}}(t) = \mathbf{1}\left[\Lambda(t) \ge \frac{1 - \pi}{\pi}\right], \qquad \Lambda(t) = \frac{p_1(t)}{p_0(t)},$$

   and connect this to Problem 1: the Bayes-optimal MIA is a likelihood-ratio test, only the threshold changes with the prior.

   (c) Suppose $T = \ell(f_\theta, z)$, the loss of model f_θ on the queried point z, and that $\ell \mid B = 1 \sim \mathcal{N}(\mu_{\mathrm{in}}, \sigma^2)$, $\ell \mid B = 0 \sim \mathcal{N}(\mu_{\mathrm{out}}, \sigma^2)$ with $\mu_{\mathrm{in}} < \mu_{\mathrm{out}}$. Show that the optimal rule reduces to a single-threshold loss test $\hat{B} = \mathbf{1}[\ell \le c]$ and compute c explicitly when $\pi = 1/2$.

Solution:

(a) By Bayes' rule,

$$\Pr[B = 1 \mid T = t] = \frac{\pi\, p_1(t)}{\pi\, p_1(t) + (1-\pi)\, p_0(t)}.$$

(b) The Bayes (0-1 loss) rule outputs $\hat{B} = 1$ iff $\Pr[B = 1 \mid T = t] \geq \Pr[B = 0 \mid T = t]$, i.e.,

$$\pi\, p_1(t) \geq (1-\pi)\, p_0(t) \iff \Lambda(t) := \frac{p_1(t)}{p_0(t)} \geq \frac{1-\pi}{\pi}.$$

By Problem 1, this is a Neyman–Pearson likelihood-ratio test at threshold $\tau = (1-\pi)/\pi$. Changing the prior π only changes the threshold; the *form* of the optimal MIA is always an LR test.

(c) **Gaussian case.** With shared variance σ^2,

$$\log \Lambda(\ell) = -\frac{(\ell - \mu_{\text{in}})^2 - (\ell - \mu_{\text{out}})^2}{2\sigma^2} = \frac{(\mu_{\text{out}} - \mu_{\text{in}})(\mu_{\text{in}} + \mu_{\text{out}} - 2\ell)}{2\sigma^2},$$

a strictly decreasing affine function of ℓ. So $\log \Lambda \geq \log((1-\pi)/\pi)$ iff

$$\ell \leq c = \frac{\mu_{\text{in}} + \mu_{\text{out}}}{2} - \frac{\sigma^2}{\mu_{\text{out}} - \mu_{\text{in}}} \log \frac{1-\pi}{\pi}.$$

For $\pi = 1/2$ the log term vanishes, giving $c = (\mu_{\text{in}} + \mu_{\text{out}})/2$.

4. **Generalization Gap as a Leakage Lower Bound (Yeom).** Let $\ell(f_\theta, z) \in [0, B]$ for all θ, z. Define

$$R_{\text{train}} = \mathbb{E}_{z \sim \text{Unif}(D)}[\ell(f_\theta, z)], \qquad R_{\text{pop}} = \mathbb{E}_{z \sim \mathcal{P}}[\ell(f_\theta, z)], \qquad \Delta = R_{\text{pop}} - R_{\text{train}} \geq 0.$$

The MIA game: sample $b \sim \text{Uniform}\{0, 1\}$. If $b = 1$, return z uniform over D; else, return $z \sim \mathcal{P}$. The adversary may use any threshold attack $\mathcal{A}_\tau(z) = \mathbf{1}[\ell(f_\theta, z) \leq \tau]$. Let $F_1(\tau) = \Pr[\ell \leq \tau \mid b = 1]$ and $F_0(\tau) = \Pr[\ell \leq \tau \mid b = 0]$.

(a) Show that the advantage of $\mathcal{A}_\tau$ equals

$$\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau).$$

(b) Using the layer-cake identity $\mathbb{E}[X] = \int_0^B (1 - F_X(t))\, \mathrm{d}t$ for $X \in [0, B]$, prove

$$\int_0^B (F_1(t) - F_0(t))\, \mathrm{d}t = \Delta.$$

(c) Conclude that there exists a threshold $\tau^\star \in [0, B]$ for which the threshold attack achieves advantage

$$\text{Adv}(\mathcal{A}_{\tau^\star}) \geq \frac{\Delta}{B}.$$

In other words, the generalization gap is a *lower* bound on the leakage of the best threshold attack — not an upper bound, contrary to a common informal slogan.
Hint: $\Delta = \int_0^B (F_1 - F_0)\, \mathrm{d}t \leq B \cdot \sup_t (F_1(t) - F_0(t))$.

(d) Now suppose additionally that under each hypothesis the loss is Gaussian:

$$\ell \mid b = 1 \sim \mathcal{N}(R_{\text{train}}, \sigma^2), \qquad \ell \mid b = 0 \sim \mathcal{N}(R_{\text{pop}}, \sigma^2).$$

Show that the maximum-advantage threshold attack has $\tau^\star = (R_{\text{train}} + R_{\text{pop}})/2$ and achieves

$$\text{Adv}^\star = 2\Phi\left(\frac{\Delta}{2\sigma}\right) - 1.$$

For small Δ/σ, derive the linear approximation $\text{Adv}^\star \approx \Delta/(\sigma\sqrt{2\pi})$ and explain why noisier loss distributions reduce leakage.

Solution:

(a) By definitions, $\text{TPR} = \Pr[\mathcal{A}_\tau = 1 \mid b = 1] = \Pr[\ell \le \tau \mid b = 1] = F_1(\tau)$ and $\text{FPR} = F_0(\tau)$. Subtracting gives $\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau)$.

(b) **Layer cake.** For a non-negative r.v. $X \in [0, B]$, by Fubini on $\{(t, X) : 0 \le t < X\}$,

$$\mathbb{E}[X] = \int_0^B \Pr[X > t]\, dt = \int_0^B (1 - F_X(t))\, dt.$$

Apply to ℓ under each hypothesis:

$$\begin{aligned} R_{\text{pop}} - R_{\text{train}} &= \int_0^B [(1 - F_0(t)) - (1 - F_1(t))]\, dt \\ &= \int_0^B (F_1(t) - F_0(t))\, dt = \Delta. \end{aligned}$$

(c) Let $h(t) = F_1(t) - F_0(t)$. If $h(t) < \Delta/B$ for every $t \in [0, B]$, then $\int_0^B h\, dt < B \cdot (\Delta/B) = \Delta$, contradicting (b). Hence some $\tau^\star \in [0, B]$ satisfies $h(\tau^\star) \ge \Delta/B$, so $\text{Adv}(\mathcal{A}_{\tau^\star}) \ge \Delta/B$. The generalization gap is a *lower* bound on the leakage of the best threshold attack, not an upper bound.

(d) **Gaussian.** With $\mu_1 = R_{\text{train}}, \mu_0 = R_{\text{pop}}$, $F_b(t) = \Phi((t - \mu_b)/\sigma)$, so

$$h(\tau) = \Phi\left(\frac{\tau - R_{\text{train}}}{\sigma}\right) - \Phi\left(\frac{\tau - R_{\text{pop}}}{\sigma}\right).$$

Setting $h'(\tau) = 0$ gives $\phi((\tau - R_{\text{train}})/\sigma) = \phi((\tau - R_{\text{pop}})/\sigma)$, so $|\tau - R_{\text{train}}| = |\tau - R_{\text{pop}}|$, i.e., $\tau^\star = (R_{\text{train}} + R_{\text{pop}})/2$. Then $\tau^\star - R_{\text{train}} = \Delta/2$, $\tau^\star - R_{\text{pop}} = -\Delta/2$, so

$$\text{Adv}^\star = \Phi(\Delta/(2\sigma)) - \Phi(-\Delta/(2\sigma)) = 2\Phi(\Delta/(2\sigma)) - 1.$$

For small $x = \Delta/(2\sigma)$, $\Phi(x) \approx 1/2 + x/\sqrt{2\pi}$, so $\text{Adv}^\star \approx \Delta/(\sigma\sqrt{2\pi})$. Larger σ (noisier loss distributions) flattens the gap and reduces leakage — the same idea behind DP noise injection.

5. **LiRA Closed Form and Decision Boundary.** Assume that for a target record z, the per-shadow-model loss has Gaussian distributions under the two membership hypotheses:

$$\ell \mid (z \in D) \sim \mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2), \qquad \ell \mid (z \notin D) \sim \mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2).$$

   (a) Derive the log-likelihood ratio

$$\log \Lambda(\ell) = \log \frac{\mathcal{N}(\ell; \mu_{\text{in}}, \sigma_{\text{in}}^2)}{\mathcal{N}(\ell; \mu_{\text{out}}, \sigma_{\text{out}}^2)} = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

   (b) Show that for general $\sigma_{\text{in}} \neq \sigma_{\text{out}}$ the optimal decision region $\{\log \Lambda > \log \tau\}$ is the complement of an interval (or a single half-line, or empty) on the loss axis. Identify when it is the complement of a bounded interval vs. a half-line.

   (c) Show that in the homoscedastic case $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$, the optimal MIA reduces to the single threshold $\mathcal{A}(z) = \mathbf{1}[\ell \leq c]$, and express c in terms of $\mu_{\text{in}}, \mu_{\text{out}}, \sigma, \tau$.

   (d) In the homoscedastic case, derive the closed-form ROC: at false-positive rate α, the optimal true-positive rate is

$$\text{TPR}(\alpha) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\text{out}} - \mu_{\text{in}}}{\sigma}\right),$$

   where Φ is the standard normal CDF.

Solution:

(a) The Gaussian log-density is $\log \mathcal{N}(\ell; \mu, \sigma^2) = -\frac{(\ell-\mu)^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$. Subtracting,

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2}{2\sigma_{\text{out}}^2} - \frac{(\ell - \mu_{\text{in}})^2}{2\sigma_{\text{in}}^2} + \log \frac{\sigma_{\text{out}}}{\sigma_{\text{in}}}.$$

(b) Rearranging, $\log \Lambda$ is quadratic in ℓ with leading coefficient

$$A = \frac{1}{2}\left(\frac{1}{\sigma_{\text{out}}^2} - \frac{1}{\sigma_{\text{in}}^2}\right).$$

- $\sigma_{\text{in}} < \sigma_{\text{out}}$ (members concentrated): $A < 0$, $\log \Lambda$ concave, $\{\log \Lambda > \log \tau\}$ is a bounded *interval* (c_-, c_+) (possibly empty, possibly a half-line if τ is extreme).
- $\sigma_{\text{in}} > \sigma_{\text{out}}$: $A > 0$, $\log \Lambda$ convex, $\{\log \Lambda > \log \tau\}$ is the *complement of a bounded interval*: $\{\ell < c_-\} \cup \{\ell > c_+\}$.
- $\sigma_{\text{in}} = \sigma_{\text{out}}$: $A = 0$, region is a half-line.

(c) **Homoscedastic.** With $\sigma_{\text{in}} = \sigma_{\text{out}} = \sigma$ the quadratic terms cancel:

$$\log \Lambda(\ell) = \frac{(\ell - \mu_{\text{out}})^2 - (\ell - \mu_{\text{in}})^2}{2\sigma^2} = \frac{(\mu_{\text{out}} - \mu_{\text{in}})(\mu_{\text{in}} + \mu_{\text{out}} - 2\ell)}{2\sigma^2},$$

a strictly decreasing affine function of ℓ (since $\mu_{\text{out}} > \mu_{\text{in}}$). Hence $\log \Lambda \geq \log \tau$ iff $\ell \leq c$, where

$$c = \frac{\mu_{\text{in}} + \mu_{\text{out}}}{2} - \frac{\sigma^2 \log \tau}{\mu_{\text{out}} - \mu_{\text{in}}}.$$

(d) $\mathrm{FPR}(c) = \Pr_{\mathrm{out}}[\ell \le c] = \Phi((c-\mu_{\mathrm{out}})/\sigma)$. Solving for c at $\mathrm{FPR} = \alpha$: $c = \mu_{\mathrm{out}} + \sigma\Phi^{-1}(\alpha)$. Substituting:

$$\mathrm{TPR}(\alpha) = \Pr_{\mathrm{in}}[\ell \le c] = \Phi\left(\frac{c-\mu_{\mathrm{in}}}{\sigma}\right) = \Phi\left(\Phi^{-1}(\alpha) + \frac{\mu_{\mathrm{out}}-\mu_{\mathrm{in}}}{\sigma}\right).$$

The discrimination is controlled entirely by the standardized gap $(\mu_{\mathrm{out}} - \mu_{\mathrm{in}})/\sigma$.

6. **Concrete MIA on a Gaussian Mean Model.** A "training" algorithm maps $D = (z_1, \ldots, z_n)$ with $z_i \in \mathbb{R}$ to the parameter

$$\theta(D) = \bar{z} + W, \quad \bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i, \quad W \sim \mathcal{N}(0, \rho^2) \text{ independent.}$$

Assume the population $\mathcal{P}$ is $\mathcal{N}(0,1)$, and the adversary observes θ plus a candidate point z^*. The MIA game: $b \sim \mathrm{Uniform}\{0,1\}$. If $b = 1$, $z^* = z_1$ (a fresh training sample, included in D); if $b = 0$, $z^* \sim \mathcal{P}$ independent of D.

(a) Derive the conditional density of θ given z^* in each hypothesis. Show that, conditional on z^*,

$$\theta \mid (b=1) \sim \mathcal{N}\left(\tfrac{z^*}{n}, \tfrac{n-1}{n^2} + \rho^2\right), \qquad \theta \mid (b=0) \sim \mathcal{N}\left(0, \tfrac{1}{n} + \rho^2\right).$$

(b) Show that the optimal adversary's test statistic is the (conditional) likelihood ratio $\Lambda(\theta, z^*) = p(\theta \mid b = 1, z^*)/p(\theta \mid b = 0, z^*)$, and write down $\log\Lambda$ as a quadratic in θ.

(c) Now suppose the inputs are clipped, $|z_i| \le 1$, so the L_2-sensitivity of $\bar{z}$ under one-element neighbors is $2/n$. The Gaussian mechanism with $\rho \ge (2/n\varepsilon)\sqrt{2\log(1.25/\delta)}$ makes the release (ε,δ)-DP. Using Problem 2(b), give a numerical upper bound on the optimal MIA advantage for $\varepsilon = 0.1$, $\delta = 10^{-5}$, and comment on why $\varepsilon = 1$ already gives a vacuous bound.

Solution:

(a) **Under $b = 0$:** z^* is drawn fresh from $\mathcal{P}$, independent of D. Since $\bar{z} = (1/n)\sum_{i=1}^{n} z_i \sim \mathcal{N}(0, 1/n)$, we have $\theta = \bar{z} + W \sim \mathcal{N}(0, 1/n + \rho^2)$, independent of z^*.

Under $b = 1$: $z^* = z_1$. Write $\bar{z} = z^*/n + (1/n)\sum_{i=2}^{n} z_i$. The sum on the right is independent of z^* with distribution $\mathcal{N}(0, (n-1)/n)$, so $(1/n)\sum_{i\ge 2} z_i \sim \mathcal{N}(0, (n-1)/n^2)$. Hence

$$\theta \mid z^* \sim \mathcal{N}\left(\frac{z^*}{n}, \frac{n-1}{n^2} + \rho^2\right).$$

(b) Let $V_1 = (n-1)/n^2 + \rho^2$ and $V_0 = 1/n + \rho^2$. Note $V_1 < V_0$ since $(n-1)/n^2 = 1/n - 1/n^2$. Then

$$\log\Lambda(\theta, z^*) = -\frac{(\theta - z^*/n)^2}{2V_1} + \frac{\theta^2}{2V_0} + \frac{1}{2}\log\frac{V_0}{V_1}.$$

Expanding,

$$\log \Lambda = \left(\frac{1}{2V_0} - \frac{1}{2V_1}\right)\theta^2 + \frac{z^*}{nV_1}\theta + \text{const}(z^*).$$

The coefficient of θ^2 is negative ($V_1 < V_0$ implies $1/V_0 < 1/V_1$), so $\log \Lambda$ is a concave parabola in θ, and the "declare member" region is an interval around the maximum.

(c) Under one-element neighbors with $|z_i|, |z_i'| \le 1$, the L_2-sensitivity of $\bar{z}$ is $|z_i - z_i'|/n \le 2/n$. The Gaussian-mechanism guarantee gives (ε, δ)-DP for

$$\rho \ge \frac{2}{n\varepsilon}\sqrt{2\log(1.25/\delta)}.$$

By Problem 2(b) the optimal MIA advantage is bounded by $\text{Adv}^\star \le e^\varepsilon - 1 + \delta$. For $\varepsilon = 0.1, \delta = 10^{-5}$: $\text{Adv}^\star \le e^{0.1} - 1 + 10^{-5} \approx 0.1052$. The optimal MIA can correctly distinguish member from non-member with advantage at most $\approx 10.5\%$ above chance.

For $\varepsilon = 1$, the bound is $e - 1 \approx 1.718 > 1$, vacuous — advantage is automatically in $[0, 1]$. Past $\varepsilon \approx \log 2 \approx 0.693$, the loose bound says nothing; the tighter Kairouz form is more informative.

7. **Influence-Function Unlearning (Closed Form).** Let $L(\theta; D) = \frac{1}{n}\sum_{i=1}^n \ell(\theta; z_i)$ be a twice-differentiable, λ-strongly convex loss in $\theta \in \mathbb{R}^d$ (so the Hessian $H_\theta = \nabla^2 L(\theta; D) \succeq \lambda I$). Let $\theta_o = \arg\min_\theta L(\theta; D)$ and $\theta_{-z} = \arg\min_\theta L(\theta; D \setminus \{z\})$.

(a) Consider the parametric family $L_t(\theta) = L(\theta; D) - \frac{t}{n}\,\ell(\theta; z)$ for $t \in [0, 1]$ and let $\theta(t) = \arg\min L_t(\theta)$ (so $\theta(0) = \theta_o$ and $\theta(1) = \theta_{-z}$ up to a normalization factor). Use the first-order optimality condition and the implicit function theorem to derive

$$\left.\frac{\mathrm{d}\theta(t)}{\mathrm{d}t}\right|_{t=0} = \frac{1}{n}\,H_{\theta_o}^{-1}\,\nabla_\theta \ell(\theta_o; z).$$

(b) Conclude the first-order Newton-step approximation

$$\theta_{-z} - \theta_o \approx \frac{1}{n}\,H_{\theta_o}^{-1}\,\nabla_\theta \ell(\theta_o; z).$$

Hence the "unlearned" parameter is $\theta_u = \theta_o + \frac{1}{n}H_{\theta_o}^{-1}\nabla_\theta \ell(\theta_o; z)$.

(c) Assume in addition that ℓ is G-Lipschitz in θ (so $\|\nabla_\theta \ell\| \le G$). Show that the L_2-sensitivity of the unlearned parameter to the choice of deleted point z versus its neighbor z' is bounded:

$$\sup_{z,z'} \|\theta_u(D, z) - \theta_u(D, z')\| \le \frac{2G}{\lambda n}.$$

(Treat $H_{\theta_o}^{-1}$ as fixed with operator norm $\le 1/\lambda$; this is the first-order sensitivity used in certified unlearning.)

Solution:

(a) **IFT derivation.** By strong convexity the unique minimizer $\theta(t) = \arg\min L_t$ exists and satisfies

$$F(\theta(t), t) := \nabla_\theta L(\theta(t); D) - \frac{t}{n}\nabla_\theta \ell(\theta(t); z) = 0.$$

Differentiating w.r.t. t:

$$\left[\nabla^2 L(\theta(t); D) - \frac{t}{n}\nabla^2 \ell(\theta(t); z)\right]\theta'(t) - \frac{1}{n}\nabla_\theta \ell(\theta(t); z) = 0.$$

At $t = 0$, the bracket equals H_{θ_o} and $\theta(0) = \theta_o$, so

$$\theta'(0) = \frac{1}{n}H_{\theta_o}^{-1}\nabla_\theta \ell(\theta_o; z).$$

($H_{\theta_o} \succeq \lambda I$ is invertible by strong convexity.)

(b) **Taylor.** First-order Taylor expansion gives

$$\theta_{-z} - \theta_o = \theta(1) - \theta(0) \approx \theta'(0) \cdot 1 = \frac{1}{n}H_{\theta_o}^{-1}\nabla_\theta \ell(\theta_o; z).$$

Hence the unlearned parameter is $\theta_u = \theta_o + \frac{1}{n}H_{\theta_o}^{-1}\nabla_\theta \ell(\theta_o; z)$, exactly the Newton-step formula in the slides.

(c) **Sensitivity.** For two candidate deleted points z, z':

$$\theta_u(D, z) - \theta_u(D, z') = \frac{1}{n}H_{\theta_o}^{-1}(\nabla \ell(\theta_o; z) - \nabla \ell(\theta_o; z')).$$

Take norms: $\|H_{\theta_o}^{-1}\|_{\text{op}} \le 1/\lambda$ by λ-strong convexity, and each gradient has norm $\le G$ by Lipschitz, so

$$\|\theta_u(D, z) - \theta_u(D, z')\| \le \frac{1}{\lambda n}(G + G) = \frac{2G}{\lambda n}.$$

This is the L_2-sensitivity needed to calibrate Gaussian noise in certified unlearning.

8. **Certified (ε, δ)-Unlearning by Gaussian Noise.** Building on Problem 7, let $A(D) = \arg\min_\theta L(\theta; D)$ be the (deterministic) retrain map, and let

$$\mathcal{U}(D, z) = \theta_o(D) + \frac{1}{n}H_{\theta_o(D)}^{-1}\nabla_\theta \ell(\theta_o(D); z) + N, \qquad N \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assume ℓ is λ-strongly convex and G-Lipschitz, and that the first-order approximation in Problem 7(b) has L_2-error bounded by $\eta := \|\theta_u^{\text{1st-order}} - A(D \setminus \{z\})\| \le C_2 G^2/(\lambda^2 n^2)$ for some universal constant C_2 (a known second-order bound; you may use this without proof).

(a) Show that the total L_2-sensitivity (relative to retrain) is bounded:

$$\left\|\mathcal{U}(D, z) - N - A(D \setminus \{z\})\right\| \le \eta \le \frac{C_2\, G^2}{\lambda^2 n^2}.$$

(b) Apply the Gaussian mechanism privacy bound: if $\sigma \geq \eta\sqrt{2\log(1.25/\delta)}/\varepsilon$, then $\mathcal{U}(D, z)$ is (ε, δ)-indistinguishable from $A(D \setminus \{z\}) + N$. Conclude

$$\sigma \geq \frac{C_2\, G^2}{\lambda^2 n^2 \varepsilon}\sqrt{2\log(1.25/\delta)}$$

gives an (ε, δ)-certified unlearning algorithm.

(c) How does the required noise scale with n? Compare to the Gaussian-mechanism noise needed for (ε, δ)-DP of the trained model itself ($\sigma \propto 1/(n\varepsilon)$). Explain in one or two sentences why the unlearning noise is smaller for large n.

Solution:

(a) **Sensitivity.** Define the first-order unlearned parameter $\theta_u^{1\text{st}} = \theta_o + \frac{1}{n}H^{-1}\nabla\ell(\theta_o; z)$, so $\mathcal{U}(D, z) = \theta_u^{1\text{st}} + N$. The exact retrain is $\theta_{-z} = A(D \setminus \{z\})$. The hypothesis bounds

$$\|\theta_u^{1\text{st}} - \theta_{-z}\| \leq \eta \leq \frac{C_2 G^2}{\lambda^2 n^2}.$$

Therefore the deterministic part of $\mathcal{U}$ differs from $A(D \setminus \{z\})$ by at most η in L_2.

(b) **Gaussian-mechanism bound.** The Gaussian mechanism: if $\|f(X) - g(X)\| \leq \eta$, then $f(X) + N$ and $g(X) + N$ (with $N \sim \mathcal{N}(0, \sigma^2 I)$) are (ε, δ)-indistinguishable when $\sigma \geq \eta\sqrt{2\log(1.25/\delta)}/\varepsilon$. Apply with $f = \theta_u^{1\text{st}}, g = \theta_{-z}$:

$$\sigma \geq \frac{\eta}{\varepsilon}\sqrt{2\log(1.25/\delta)} \quad \Longleftarrow \quad \sigma \geq \frac{C_2 G^2}{\lambda^2 n^2 \varepsilon}\sqrt{2\log(1.25/\delta)},$$

matching the Guo et al. (2020) bound.

(c) Unlearning noise scales as $\sigma \propto 1/n^2$, whereas DP-of-the-trained-model noise scales as $\sigma_{\text{DP}} \propto 1/n$. The unlearning task is *easier* for large n: a single training point's influence is itself $O(1/n)$, and the Newton correction reduces the residual to $O(1/n^2)$ — much smaller than what full DP would need to mask.

9. **NPO Has a Bounded Gradient; GA Does Not.** For LLM unlearning, let $\pi_\theta(y \mid x)$ denote the model's likelihood of completion y given prompt x, and let π_{ref} be a fixed reference model. The forget set is $\mathcal{D}_F$. Two unlearning losses:

$$\begin{aligned}
\mathcal{L}_{\text{GA}}(\theta) &= -\,\mathbb{E}_{(x,y)\sim\mathcal{D}_F}[-\log \pi_\theta(y \mid x)] = \mathbb{E}[\log \pi_\theta(y \mid x)]\,,\\
\mathcal{L}_{\text{NPO}}(\theta) &= \tfrac{2}{\beta}\,\mathbb{E}\Big[\log\Big(1 + \big(\tfrac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\big)^\beta\Big)\Big] \qquad (\beta > 0).
\end{aligned}$$

(a) Compute $\nabla_\theta \mathcal{L}_{\text{GA}}$. Using the identity $\nabla_\theta \log \pi_\theta = \nabla_\theta \pi_\theta / \pi_\theta$, show that for any sample with $\pi_\theta(y \mid x) \to 0$ (and $\|\nabla_\theta \pi_\theta\|$ bounded away from zero in some direction), $\|\nabla_\theta \mathcal{L}_{\text{GA}}\| \to \infty$. Explain in one sentence why this drives the GA optimization to escape the data distribution ("GA collapse").

(b) Let $u(\theta) = \beta \log(\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x))$. Show that

$$\mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta}\,\mathbb{E}[\log(1 + e^{u(\theta)})], \qquad \frac{\partial \mathcal{L}_{\text{NPO}}}{\partial u} = \frac{2}{\beta}\,\sigma(u) \in \left(0, \tfrac{2}{\beta}\right),$$

where σ is the logistic sigmoid. Conclude that

$$\big\|\nabla_\theta \mathcal{L}_{\text{NPO}}\big\| \le 2\,\mathbb{E}\big[\|\nabla_\theta \log \pi_\theta(y \mid x)\|\big].$$

(c) Show that this gradient bound is uniform in θ (it does not blow up as $\pi_\theta \to 0$), in sharp contrast to part (a). State briefly why this bounded-per-step divergence is what makes NPO "safe" for unlearning.

Solution:

(a) **GA gradient blowup.** By definition

$$\nabla_\theta \mathcal{L}_{\text{GA}}(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{D}_F}[\nabla_\theta \log \pi_\theta(y \mid x)] = \mathbb{E}\left[\frac{\nabla_\theta \pi_\theta(y \mid x)}{\pi_\theta(y \mid x)}\right].$$

If a forget sample has $\pi_\theta(y \mid x) \to 0$ with $\|\nabla_\theta \pi_\theta(y \mid x)\|$ bounded away from 0 in some direction, the integrand blows up. Therefore $\|\nabla_\theta \mathcal{L}_{\text{GA}}\| \to \infty$, and a single gradient-ascent step takes an arbitrarily large jump in θ, pushing the model off the data manifold and destroying fluency on *unrelated* inputs — the "GA collapse" observed by Jang et al.

(b) **NPO bounded gradient.** Let $u(\theta) = \beta \log(\pi_\theta(y \mid x)/\pi_{\text{ref}}(y \mid x))$. Then

$$\mathcal{L}_{\text{NPO}}(\theta) = \frac{2}{\beta}\mathbb{E}[\log(1 + e^{u(\theta)})], \quad \frac{\partial \mathcal{L}_{\text{NPO}}}{\partial u} = \frac{2}{\beta} \cdot \frac{e^u}{1 + e^u} = \frac{2}{\beta}\sigma(u),$$

with $\sigma(u) \in (0, 1)$, so $\partial \mathcal{L}_{\text{NPO}}/\partial u \in (0, 2/\beta)$. By the chain rule,

$$\nabla_\theta \mathcal{L}_{\text{NPO}} = \mathbb{E}\left[\frac{\partial \mathcal{L}}{\partial u}\nabla_\theta u\right] = \mathbb{E}\left[\frac{2}{\beta}\sigma(u) \cdot \beta \nabla_\theta \log \pi_\theta\right] = 2\mathbb{E}[\sigma(u)\nabla_\theta \log \pi_\theta],$$

hence $\|\nabla_\theta \mathcal{L}_{\text{NPO}}\| \le 2\mathbb{E}[\|\nabla_\theta \log \pi_\theta\|]$.

(c) As unlearning progresses, $\pi_\theta(y \mid x) \to 0$ on forget samples, so $u \to -\infty$ and $\sigma(u) \to 0$, shrinking the per-sample contribution even though $\nabla_\theta \log \pi_\theta$ itself blows up like $1/\pi_\theta$. More precisely, $\sigma(u) \sim e^u = (\pi_\theta/\pi_{\text{ref}})^\beta$ for $u \to -\infty$, so

$$\sigma(u)\nabla_\theta \log \pi_\theta \sim \frac{\pi_\theta^{\beta-1}}{\pi_{\text{ref}}^{\beta}}\nabla_\theta \pi_\theta,$$

which stays bounded for $\beta \ge 1$ (and remains tame in practice for smaller β). Operationally: once a forget sample's likelihood is small enough, NPO essentially stops updating θ on it. GA never stops, which is why it collapses; NPO's sigmoid saturation is the entire fix.

10. **Black-Box Unlearning Audits Require Many Queries.** Let θ_o and θ_u be two LLMs with output distributions P_0 and P_1 on prompts $x \in \mathcal{X}$. Let $\mathrm{TV}(P_0, P_1) = \frac{1}{2}\int |p_0(y \mid x) - p_1(y \mid x)|\, \mathrm{d}\mu(y, x) \leq \eta$ (averaged over a fixed query distribution). An auditor draws q i.i.d. prompts $x_1, \ldots, x_q$, observes one sample $y_i \sim P_b(\cdot \mid x_i)$ per prompt, and outputs $\hat{b} \in \{0, 1\}$.

    (a) **(Le Cam two-point lower bound.)** Show that for any test based on q samples,

    $$\Pr_{b=0}[\hat{b} \neq 0] + \Pr_{b=1}[\hat{b} \neq 1] \geq 1 - \mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}).$$

    Hint: For any test ψ and any pair of distributions P, Q, $\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi] \leq \mathrm{TV}(P, Q)$.

    (b) Show that $\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \leq q \cdot \mathrm{TV}(P_0, P_1) \leq q\eta$ by a hybrid (telescoping) argument over the q coordinates.

    (c) Conclude that to guarantee total error $\leq \alpha$ (i.e., $\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \leq 2\alpha$), the auditor needs

    $$q \geq \frac{1 - 2\alpha}{\eta}.$$

    Tighter χ^2-style bounds give $q = \Omega(1/\eta^2)$. Why does this make single-prompt audits of LLM unlearning fundamentally insufficient when θ_u, θ_o differ only by a small distributional perturbation?

    Solution:

    (a) **Le Cam.** Identify $\hat{b}$ with a test $\psi \in [0, 1]$ (the conditional probability of declaring $b = 1$ given the observations). With $P = P_1^{\otimes q}$ and $Q = P_0^{\otimes q}$,

    $$\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] = \mathbb{E}_Q[\psi] + \mathbb{E}_P[1 - \psi] = 1 - (\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi]).$$

    By the variational definition of TV, $\mathbb{E}_P[\psi] - \mathbb{E}_Q[\psi] \leq \mathrm{TV}(P, Q)$. Therefore

    $$\Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \geq 1 - \mathrm{TV}(P_1^{\otimes q}, P_0^{\otimes q}).$$

    (b) **Hybrid (telescoping).** Construct intermediate distributions $H_k = P_0^{\otimes k} \otimes P_1^{\otimes (q-k)}$ for $k = 0, \ldots, q$, so $H_q = P_0^{\otimes q}$ and $H_0 = P_1^{\otimes q}$. By the triangle inequality for TV,

    $$\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \leq \sum_{k=1}^{q} \mathrm{TV}(H_k, H_{k-1}).$$

    Each pair H_k, H_{k-1} differs only in coordinate k: H_k has P_0 there, H_{k-1} has P_1. By the product structure, $\mathrm{TV}(H_k, H_{k-1}) = \mathrm{TV}(P_0, P_1) \leq \eta$. Summing, $\mathrm{TV}(P_0^{\otimes q}, P_1^{\otimes q}) \leq q\eta$.

(c) Combining (a) and (b):

$$2\alpha \geq \Pr_0[\hat{b} \neq 0] + \Pr_1[\hat{b} \neq 1] \geq 1 - q\eta \implies q \geq \frac{1-2\alpha}{\eta}.$$

For $\alpha = 0.1$ this needs $q \geq 0.8/\eta$. Tighter Pinsker/Hellinger bounds with $\mathrm{TV}(P^{\otimes q}, Q^{\otimes q})^2 \leq 2q\,\mathrm{KL}(P\|Q)$ give the well-known $q = \Omega(1/\eta^2)$ scaling (matching χ^2 form behind Pawelczyk et al. 2024).

Implication for unlearning: if θ_u and θ_o differ in output distribution by a small TV η, a black-box auditor with sub-$1/\eta$ queries cannot reliably distinguish them. Single-prompt audits — the common regulatory-checkbox approach — fundamentally cannot certify that unlearning has happened; the auditor must use distributional probes.